

Enhanced Convex Relaxation Approach for Co-operated Power and Water Systems

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Abstract—This paper introduces a novel framework for optimizing the cooperative operation of water and power systems, specifically incorporating variable speed pumps. To tackle the non-convex, nonlinear constraints in optimal water flow and pump scheduling, we propose a direct relaxation method that transforms the problem into a convex form, complemented by an iterative bound-tightening algorithm. This approach significantly simplifies the original problem, demonstrating superior computational efficiency and providing reliable near-optimal solutions compared to traditional approximation methods. Through comparative analysis, our method proves to be a practical and scalable solution for large-scale industrial systems, offering effective options for real-world implementation.

Index Terms—power distribution system, variable speed pump, convex relaxation, optimization

NOMENCLATURE

Sets

$\mathcal{T}$	ordered set of hours within optimization horizon
$\mathcal{G}$	set of generating units
$\mathcal{G}(m)$	set of generating units that are connected to bus m
$\mathcal{M}$	set of electricity-system buses
$\mathcal{M}(m)$	set of buses that are connected to bus m
$\mathcal{I}, \mathcal{I}^J, \mathcal{I}^R$	set of water-system all, junction and reservoir nodes
$\mathcal{I}^+(i)$	set of nodes that flow water to node i
$\mathcal{I}^-(i)$	set of nodes that flow water directly from node i
$\mathcal{I}^{L,+}(l)$	node that is the inlet of pipe l
$\mathcal{I}^{L,-}(l)$	node that is the outlet of pipe l
$\mathcal{I}^{K,+}(k)$	node that is the inlet of tank k
$\mathcal{I}^{K,-}(k)$	node that is the outlet of tank k
$\mathcal{I}^{P,+}(p)$	node that is the inlet of pump p
$\mathcal{I}^{P,-}(p)$	node that is the outlet of pump p
$\mathcal{I}^{V,+}(v)$	node that is the inlet of valve v
$\mathcal{I}^{V,-}(v)$	node that is the outlet of valve v
$\mathcal{L}, \mathcal{K}, \mathcal{P}, \mathcal{V}$	sets of water pipes, tanks, pumps and valves
$\mathcal{P}(m)$	set of water pumps that are connected to bus m
Ω	set of convexification tightening iteration numbers

Parameters

$C_{m,t}^E$	bus- m /hour- t value of lost electric load (\$/MW)
$C_{g,t}^G$	generator- g /hour- t operating cost (\$/MW)
$C_{i,t}^W$	node- i /hour- t value of lost water load (\$/m ³)
$\bar{D}_{m,t}^E$	bus- m /hour- t electricity demand (MW)
$\bar{D}_{i,t}^W$	node- i /hour- t water demand (m ³ /s)
$\bar{F}_{m,n}$	capacity of the transmission line (MW)

$G_g^{ramp,+}$	ramp-up limit of generating unit g (MW/h)
$G_g^{ramp,-}$	ramp-down limit of generating unit g (MW/h)
$\bar{G}_g$	capacity of generating unit g (MW)
$\bar{H}_i, H_i^-$	elevation and minimum pressure head of node i (m)
K_k^A	cross-sectional area of tank k (m ²)
$\bar{K}_k$	capacity of tank k (m ³)
L_l^p, L_l^L	inner radius and length of the pipe l (m)
L_l^Δ	Darcy friction parameter of the pipe l
P_p^Q	pump p head gain water-flow coefficient (s ² /m ⁵)
$P_p^{Q,\sigma}$	head gain bilinear coefficient of pump p (s/m ²)
P_p^σ	head gain pump-speed coefficient on pump p (m)
$P_p^\eta, \bar{P}_p^\sigma$	efficiency and maximum speed of pump p (p.u.)
$U_{m,n}$	susceptance buses m and n (MW/rad)
γ	gravitational constant (m/s ²)

Decision Variables

C_ω	water and power system total operation cost (\$)
$D_{m,t}^E$	served bus- m /hour- t electricity demand (MW)
$D_{i,t}^W$	served node- i /hour- t water demand (m ³ /s)
$F_{m,n,t}$	hour- t power flow from bus m to n (MW)
$G_{g,t}$	hour- t output of generating unit g (MW)
$H_{i,t}$	hour- t head pressure of node i (m)
$H_{l,t}^L, H_{v,t}^V$	hour- t head loss of pipe l and valve v (m)
$H_{p,t}^P$	hour- t head gain of pump p (m)
$P_{p,t}^E$	hour- t electricity consumption of pump p (MW)
$P_{p,t}$	hour- t speed of the pump p (p.u.)
$Q_{i,j,t}$	hour- t water flow between nodes i and j (m ³ /s)
$\theta_{m,t}$	bus- m /hour- t phase angle (rad)

I. INTRODUCTION

Amid rising energy and water demand, optimizing water distribution network (WDN) and power distribution network (PDN) operations is vital, as electricity for pumping constitutes over 90% of operational expenses for many utilities, and water systems consume approximately 4% of national electricity usage in the U.S. [1]. The cooperative optimization of water and power systems (C-OWPF) offers significant potential for cost reduction, operational flexibility, and sustainability [2]–[4]. These optimization problems are often modeled as nonlinear programming (NLP) due to the complex water flow dynamics, which pose significant computational challenges, especially for variable-speed pumps (VSPs).

Previous approaches most focus on simplifying models through approximation methods for fixed-speed pumps (FSPs) [3], [5]. Verleye et al. [6] introduced a heuristic using piecewise linearization, which has been widely adopted in WDN applications [7], [8]. Extensions of this linearization technique are also studied on VSP dynamics [9], but these methods increase model complexity by adding binary variables. Two works proposed convexification approaches for VSPs, but they all involve multi-step procedures, limiting their practicality and scalability [4], [10].

In this paper, we propose a direct relaxation method to simplify WDN optimization involving VSPs. By converting nonconvex constraints into convex envelopes and incorporating an iterative bound-tightening algorithm, our approach enhances computational efficiency and extends applicability. Comparative analysis shows significant improvements in feasibility gaps, making our method a promising alternative for real-world implementations.

II. MODEL FORMULATIONS

In this section, we introduce a model for the optimal coordination of WDN and PDN. Our objective is to minimize the total operational cost subject to the constraints of both systems. The model is formulated as:

$$\begin{aligned} \min \quad & \sum_{g \in \mathcal{G}, t \in \mathcal{T}} C_{g,t}^G \cdot G_{g,t} + \sum_{m \in \mathcal{M}, t \in \mathcal{T}} C_{m,t}^E \cdot (\bar{D}_{m,t}^E - D_{m,t}^E) \\ & + \sum_{i \in \mathcal{I}^J, t \in \mathcal{T}} C_{i,t}^W \cdot (\bar{D}_{i,t}^W - D_{i,t}^W) \end{aligned} \quad (1)$$

$$\text{s.t.} \quad \sum_{j \in \mathcal{I}^+(i)} Q_{j,i,t} - \sum_{j \in \mathcal{I}^-(i)} Q_{i,j,t} = D_{i,t}^W, \forall i \in \mathcal{I}, t \in \mathcal{T} \quad (2)$$

$$\begin{aligned} K_k^A \cdot (H_{i,t} - H_{i,t-1}) &= \sum_{z \in \mathcal{I}^+(j)} Q_{z,j,t} - \sum_{z \in \mathcal{I}^-(i)} Q_{i,z,t}; \\ \forall k \in \mathcal{K}, i \in \mathcal{I}^{K,-}(k), j \in \mathcal{I}^{K,+}(k), t \in \mathcal{T}; \end{aligned} \quad (3)$$

$$H_{j,t} \leq H_{i,t}; \forall k \in \mathcal{K}, i \in \mathcal{I}^{K,+}(k), j \in \mathcal{I}^{K,-}(k), t \in \mathcal{T} \quad (4)$$

$$0 \leq K_k^A \cdot H_{i,t} \leq \bar{K}_k; \forall k \in \mathcal{K}, i \in \mathcal{I}^{K,-}(k), t \in \mathcal{T}; \quad (5)$$

$$\begin{aligned} P_p^Q \cdot Q_{i,j,t}^2 + P_p^{Q,\sigma} \cdot Q_{i,j,t} \cdot P_{p,t} + P_p^\sigma \cdot P_{p,t}^2 &= H_{p,t}^P \\ \forall p \in \mathcal{P}, i \in \mathcal{I}^{P,+}(p), j \in \mathcal{I}^{P,-}(p), t \in \mathcal{T}; \end{aligned} \quad (6)$$

$$\begin{aligned} \gamma \cdot H_{p,t}^P \cdot Q_{i,j,t} &= P_p^\eta \cdot P_{p,t}^E; \\ \forall p \in \mathcal{P}, i \in \mathcal{I}^{P,+}(p), j \in \mathcal{I}^{P,-}(p), t \in \mathcal{T}; \end{aligned} \quad (7)$$

$$\begin{aligned} L_l^\Delta \cdot L_l^L \cdot Q_{i,j,t}^2 &= 4 \cdot \pi^2 \cdot (L_l^P)^5 \cdot \gamma \cdot H_{l,t}^L; \\ \forall l \in \mathcal{L}, i \in \mathcal{I}^{L,+}(l), j \in \mathcal{I}^{L,-}(l), t \in \mathcal{T}; \end{aligned} \quad (8)$$

$$\begin{aligned} H_{j,t} - H_{i,t} + \bar{H}_j - \bar{H}_i &= H_{p,t}^P; \\ \forall p \in \mathcal{P}, i \in \mathcal{I}^{P,+}(p), j \in \mathcal{I}^{P,-}(p), t \in \mathcal{T}; \end{aligned} \quad (9)$$

$$\begin{aligned} H_{i,t} - H_{j,t} + \bar{H}_i - \bar{H}_j &= L_{i,j,t} H_{l,t}^L; \\ \forall l \in \mathcal{L}, i \in \mathcal{I}^{L,+}(l), j \in \mathcal{I}^{L,-}(l), t \in \mathcal{T}; \end{aligned} \quad (10)$$

$$\begin{aligned} H_{i,t} - H_{j,t} + \bar{H}_i - \bar{H}_j &= H_{l,t}^V; \\ \forall v \in \mathcal{V}, i \in \mathcal{I}^{V,+}(v), j \in \mathcal{I}^{V,-}(v), t \in \mathcal{T}; \end{aligned} \quad (11)$$

$$H_{i,t} = 0; \forall i \in \mathcal{I}^R, t \in \mathcal{T}; \quad (12)$$

$$H_i^- \leq H_{i,t}; \forall i \in \mathcal{I}^J, t \in \mathcal{T}; \quad (13)$$

$$0 \leq D_{i,t}^W \leq \bar{D}_{i,t}^W; \forall i \in \mathcal{I}^J, t \in \mathcal{T}; \quad (14)$$

$$0 \leq Q_{i,j,t}; \forall i \in \mathcal{I}, j \in \mathcal{I}^-(i), t \in \mathcal{T}; \quad (15)$$

$$0 \leq P_{p,t} \leq \bar{P}_p^\sigma; \forall p \in \mathcal{P}, t \in \mathcal{T}; \quad (16)$$

$$\begin{aligned} \sum_{g \in \mathcal{G}(m)} G_{g,t} + \sum_{n \in \mathcal{M}(m)} F_{m,n,t} \\ = D_{m,t}^E + \sum_{p \in \mathcal{P}(m)} P_{p,t}^E; \forall m \in \mathcal{M}, t \in \mathcal{T}; \end{aligned} \quad (17)$$

$$\begin{aligned} F_{m,n,t} &= U_{m,n} (\theta_{m,t} - \theta_{n,t}); \\ \forall m \in \mathcal{M}, n \in \mathcal{M}(m), t \in \mathcal{T}; \end{aligned} \quad (18)$$

$$\begin{aligned} -\bar{F}_{m,n,t} \leq F_{m,n,t} \leq \bar{F}_{m,n,t}; \\ \forall m \in \mathcal{M}, n \in \mathcal{M}(m), t \in \mathcal{T}; \end{aligned} \quad (19)$$

$$\theta_{REF,t} = 0; \forall t \in \mathcal{T}; \quad (20)$$

$$-\pi \leq \theta_{m,t} \leq \pi; \forall m \in \mathcal{M}, t \in \mathcal{T}; \quad (21)$$

$$0 \leq D_{m,t}^E \leq \bar{D}_{m,t}^E; \forall m \in \mathcal{M}, t \in \mathcal{T}; \quad (22)$$

$$0 \leq G_{g,t} \leq \bar{G}_{g,t}; \forall g \in \mathcal{G}, t \in \mathcal{T}; \quad (23)$$

$$\begin{aligned} -G_g^{ramp,-} \leq G_{g,t} - G_{g,t-1} \leq G_g^{ramp,+}; \\ \forall g \in \mathcal{G}, t \in \mathcal{T}; \end{aligned} \quad (24)$$

Objective function (1) minimizes the total operation cost of the cooperated WDS and PDS. The first term in the objective function gives the cost of power generation, while the second and third are the cost of unserved electric and water demand. Constraints (2) ensures the hourly balance of water flow. Constraints (3)–(5) account for dynamic water levels of the water tanks, limit the outlet head pressure to be less than the inlet and enforced the tank's capacity. Constraints (6) and (7) impose the increase in water pressure of pumps and the corresponding energy consumption [11]. Constraints (8) are described by the Darcy–Weisbach frictional head loss equations along pipes [12]. Constraints (9)–(11) enforce that the nodes head pressure difference. Constraints (12) set the pressure head of reservoirs to zero. Constraints (13) impose lower pressure head bounds of junctions. Constraints (14) ensure that the served water demands bounds. Constraints (15) enforce water flow directions. Constraints (16) impose maximum pump speed. Constraints (17) ensure power flow balance. Constraints (18) and (19) impose power flow limits. Constraints (20) fix the reference bus phase angle to zero while constraints (21) enforce upper and lower phase angle bounds. Constraints (22) provide bounds for served demand. Constraints (23) and (24) impose the capacity and ramping limits of the generators.

III. COMPUTATION

A. Convex Relaxation of Water Distribution Model

The integrated model, represented by equations (1)–(24), is identified as a non-convex non-linear optimization problem. This complexity is from the equations describing the water

pumps' head gain (6), the energy consumption of these pumps (7), and the head loss in pipes due to friction (8).

For any given iteration ω , there always exist upper and lower bounds, denoted as $Q_{i,j,t,\omega}^+$, $Q_{i,j,t,\omega}^-$ for water flow $Q_{i,j,t}$, and $P_{p,t,\omega}^+$, $P_{p,t,\omega}^-$ for pump speed variables $P_{p,t}$.

Constraint (8) can be equivalently written as:

$$L_l^\Delta \cdot L_l^L \cdot Q_{i,j,t}^2 \geq 4 \cdot \pi^2 \cdot (L_l^p)^5 \cdot \gamma \cdot H_{l,t}^L; \quad \forall l \in \mathcal{L}, i \in \mathcal{I}^{L,+}(l), j \in \mathcal{I}^{L,-}(l), t \in \mathcal{T}; \quad (25)$$

$$L_l^\Delta \cdot L_l^L \cdot Q_{i,j,t}^2 \leq 4 \cdot \pi^2 \cdot (L_l^p)^5 \cdot \gamma \cdot H_{l,t}^L; \quad \forall l \in \mathcal{L}, i \in \mathcal{I}^{L,+}(l), j \in \mathcal{I}^{L,-}(l), t \in \mathcal{T}; \quad (26)$$

Noted that constraint (25) inherently possesses convex properties. Conversely, constraint (26) is concave, permitting a relaxation to a linear form that crosses the original function on $Q_{i,j,t,\omega}^+$, $Q_{i,j,t,\omega}^-$ as constraint (27).

$$L_l^\Delta \cdot L_l^L \cdot [(Q_{i,j,t,\omega}^+ + Q_{i,j,t,\omega}^-) \cdot Q_{i,j,t} - Q_{i,j,t,\omega}^+ \cdot Q_{i,j,t,\omega}^-] \leq 4 \cdot \pi^2 \cdot (L_l^p)^5 \cdot \gamma \cdot H_{l,t}^L; \quad \forall l \in \mathcal{L}, i \in \mathcal{I}^{L,+}(l), j \in \mathcal{I}^{L,-}(l), t \in \mathcal{T}; \quad (27)$$

According to pump characteristics [11], where P_p^Q is negative and both $P_p^{Q,\sigma}$ and P_p^σ are positive, for any given positive $Q_{i,j,t}$, the function (8) is continuous and monotonically increasing with $P_{p,t}$. So, given the defined bounds on pump speed, the constraints can be equally transformed to:

$$P_p^Q \cdot Q_{i,j,t}^2 + P_p^{Q,\sigma} \cdot Q_{i,j,t} \cdot P_{p,t,\omega}^+ + P_p^\sigma \cdot P_{p,t,\omega}^+{}^2 \geq H_{p,t}^P; \quad \forall p \in \mathcal{P}, i \in \mathcal{I}^{P,+}(p), j \in \mathcal{I}^{P,-}(p), t \in \mathcal{T}; \quad (28)$$

$$P_p^Q \cdot Q_{i,j,t}^2 + P_p^{Q,\sigma} \cdot Q_{i,j,t} \cdot P_{p,t,\omega}^- + P_p^\sigma \cdot P_{p,t,\omega}^-{}^2 \leq H_{p,t}^P; \quad \forall p \in \mathcal{P}, i \in \mathcal{I}^{P,+}(p), j \in \mathcal{I}^{P,-}(p), t \in \mathcal{T}; \quad (29)$$

Constraint (29) is affirmed as convex. The concave counterpart, constraint (28), can subsequently relax to the linear lower bound (30).

$$(2P_p^Q \cdot Q_{i,j,t,\omega}^- + P_p^{Q,\sigma} \cdot P_{p,t,\omega}^+) \cdot Q_{i,j,t} + P_p^\sigma \cdot P_{p,t,\omega}^+{}^2 - P_p^Q \cdot Q_{i,j,t,\omega}^+{}^2 \geq H_{p,t}^P; \quad \forall p \in \mathcal{P}, i \in \mathcal{I}^{P,+}(p), j \in \mathcal{I}^{P,-}(p), t \in \mathcal{T}; \quad (30)$$

The function (7) is also continuous and monotonically increasing with $P_{p,t}$. So we can refine the pumps' energy consumption constraints based on the upper and lower bounds of pump speed, given as:

$$P_p^Q \cdot Q_{i,j,t}^3 + P_p^{Q,\sigma} \cdot Q_{i,j,t}^2 \cdot P_{p,t,\omega}^+ + P_p^\sigma \cdot Q_{i,j,t} \cdot P_{p,t,\omega}^+{}^2 \geq P_p^\eta \cdot P_{p,t,\omega}^E / \gamma; \quad \forall p \in \mathcal{P}, i \in \mathcal{I}^{P,+}(p), j \in \mathcal{I}^{P,-}(p), t \in \mathcal{T}; \quad (31)$$

$$P_p^Q \cdot Q_{i,j,t}^3 + P_p^{Q,\sigma} \cdot Q_{i,j,t}^2 \cdot P_{p,t,\omega}^- + P_p^\sigma \cdot Q_{i,j,t} \cdot P_{p,t,\omega}^-{}^2 \leq P_p^\eta \cdot P_{p,t,\omega}^E / \gamma; \quad \forall p \in \mathcal{P}, i \in \mathcal{I}^{P,+}(p), j \in \mathcal{I}^{P,-}(p), t \in \mathcal{T}; \quad (32)$$

As introduced by [13], these cubic constraints have a convex envelope method.

For the constraint (31), when $Q_{i,j,t,\omega}^- \leq a_{p,t,\omega}^{1,+} = -P_p^{Q,\sigma} P_{p,t,\omega}^+ / 3P_p^Q$, the constraint is non-convex form. For a given $Q_{i,j,t,\omega}^+$, when the $Q_{i,j,t,\omega}^+ = a_{p,t,\omega}^{2,+} = -0.5Q_{i,j,t,\omega}^- - P_p^{Q,\sigma} P_{p,t,\omega}^+ / 2P_p^Q$, then the line connecting these two points intersects the function (31) at only those two points and is tangent at $Q_{i,j,t,\omega}^+$. Thus, when $Q_{i,j,t,\omega}^- \leq a_{p,t,\omega}^{1,+}$ and $Q_{i,j,t,\omega}^+ \leq a_{p,t,\omega}^{2,+}$, the non-convex constraint (31) can be relaxed to a linear constraint that across through the water flow lower and upper bounds as:

$$[P_p^Q (Q_{i,j,t,\omega}^+{}^2 + Q_{i,j,t,\omega}^+ \cdot Q_{i,j,t,\omega}^- + Q_{i,j,t,\omega}^-{}^2) + P_p^{Q,\sigma} \cdot P_{p,t,\omega}^+ (Q_{i,j,t,\omega}^+ + Q_{i,j,t,\omega}^-) + P_p^\sigma \cdot P_{p,t,\omega}^+{}^2] \cdot Q_{i,j,t} - P_p^Q \cdot Q_{i,j,t,\omega}^+ \cdot Q_{i,j,t,\omega}^- (Q_{i,j,t,\omega}^+ + Q_{i,j,t,\omega}^-) - P_p^{Q,\sigma} \cdot P_{p,t,\omega}^+ \cdot Q_{i,j,t,\omega}^+ \cdot Q_{i,j,t,\omega}^- \geq P_p^\eta \cdot P_{p,t,\omega}^E / \gamma; \quad \forall p \in \mathcal{P}, i \in \mathcal{I}^{P,+}(p), j \in \mathcal{I}^{P,-}(p), t \in \mathcal{T}; \quad (33)$$

When $Q_{i,j,t,\omega}^- \leq a_{p,t,\omega}^{1,+}$ and $Q_{i,j,t,\omega}^+ > a_{p,t,\omega}^{2,+}$, as figure 1 shown, the constraint (31) can be replaced by the linear constraint (34) pass the lower bound water flow point and tangent with the constraint on the point $Q_{i,j,t} = a_{p,t,\omega}^{2,+}$, as well as the linear constraint (35) that tangent with the cubic function on the water flow upper bound point.

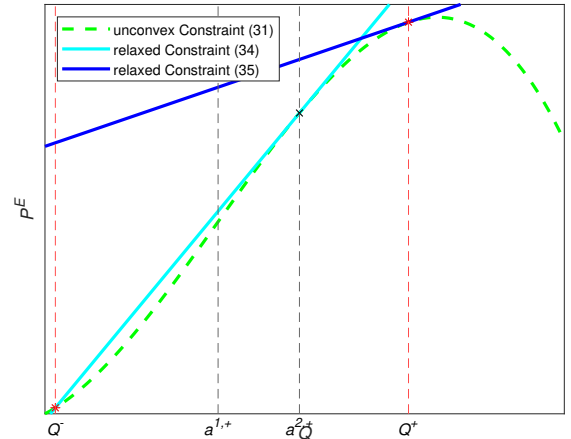


Fig. 1. Convexified approximation of water pump energy consumption function when $Q_{i,j,t,\omega}^- \leq a_{p,t,\omega}^{1,+}$ and $Q_{i,j,t,\omega}^+ > a_{p,t,\omega}^{2,+}$.

$$(0.75P_p^Q \cdot Q_{i,j,t,\omega}^-{}^2 + 0.5P_p^{Q,\sigma} \cdot Q_{i,j,t,\omega}^- \cdot P_{p,t,\omega}^+ - 0.25P_p^\sigma \cdot P_{p,t,\omega}^+{}^2 / P_p^Q + P_p^\sigma \cdot P_{p,t,\omega}^+{}^2) \cdot Q_{i,j,t} + 0.25P_p^Q \cdot Q_{i,j,t,\omega}^-{}^3 + 0.5P_p^{Q,\sigma} \cdot Q_{i,j,t,\omega}^-{}^2 \cdot P_{p,t,\omega}^+ + 0.5P_p^{Q,\sigma} \cdot Q_{i,j,t,\omega}^- \cdot P_{p,t,\omega}^+{}^2 / P_p^Q \geq P_p^\eta \cdot P_{p,t,\omega}^E / \gamma; \quad \forall p \in \mathcal{P}, i \in \mathcal{I}^{P,+}(p), j \in \mathcal{I}^{P,-}(p), t \in \mathcal{T}; \quad (34)$$

$$(3P_p^Q \cdot Q_{i,j,t,\omega}^+{}^2 + 2P_p^{Q,\sigma} \cdot Q_{i,j,t,\omega}^+ \cdot P_{p,t,\omega}^+ + P_p^\sigma \cdot P_{p,t,\omega}^+{}^2) \cdot Q_{i,j,t} - 2P_p^Q \cdot Q_{i,j,t,\omega}^+{}^3 - P_p^{Q,\sigma} \cdot Q_{i,j,t,\omega}^+{}^2 \cdot P_{p,t,\omega}^+ \geq P_p^\eta \cdot P_{p,t,\omega}^E / \gamma; \quad \forall p \in \mathcal{P}, i \in \mathcal{I}^{P,+}(p), j \in \mathcal{I}^{P,-}(p), t \in \mathcal{T}; \quad (35)$$

When $Q_{i,j,t,\omega}^- > a_{p,t,\omega}^{1,+}$ the constraint (31) is convex.

Similarly, for constraint (32), when $Q_{i,j,t,\omega}^+ \leq a_{p,t,\omega}^{1,-}$, it is convex. When $Q_{i,j,t,\omega}^+ > a_{p,t,\omega}^{1,-}$ and $Q_{i,j,t,\omega}^- \leq a_{p,t,\omega}^{2,-} = -0.5Q_{i,j,t,\omega}^+ - P_p^{Q,\sigma}P_{p,t,\omega}^-/2P_p^Q$, the constraint (32) shift to non-convex and can relax to:

$$\begin{aligned} & (0.75P_p^Q \cdot Q_{i,j,t,\omega}^+)^2 + 0.5P_p^{Q,\sigma} \cdot Q_{i,j,t,\omega}^+ \cdot P_{p,t,\omega}^- - \\ & 0.25P_p^\sigma \cdot P_p^{Q,\sigma^2} \cdot P_{p,t,\omega}^2 / P_p^Q + P_p^\sigma \cdot P_{p,t,\omega}^2 \cdot Q_{i,j,t} \\ & + 0.25P_p^Q \cdot Q_{i,j,t,\omega}^+{}^3 + 0.5P_p^{Q,\sigma} \cdot Q_{i,j,t,\omega}^+{}^2 \cdot P_{p,t,\omega}^- \\ & + 0.5P_p^{Q,\sigma^2} \cdot Q_{i,j,t,\omega}^+ \cdot P_{p,t,\omega}^2 / P_p^Q \leq P_p^\eta \cdot P_{p,t}^E / \gamma; \\ & \forall p \in \mathcal{P}, i \in \mathcal{I}^{P,+}(p), j \in \mathcal{I}^{P,-}(p), t \in \mathcal{T}; \quad (36) \\ & (3P_p^Q \cdot Q_{i,j,t,\omega}^-{}^2 + 2P_p^{Q,\sigma} \cdot Q_{i,j,t,\omega}^- \cdot P_{p,t,\omega}^- + P_p^\sigma \cdot P_{p,t,\omega}^-{}^2) \cdot \\ & Q_{i,j,t} - 2P_p^Q \cdot Q_{i,j,t,\omega}^-{}^3 - P_p^{Q,\sigma} \cdot Q_{i,j,t,\omega}^-{}^2 \cdot P_{p,t,\omega}^- \leq P_p^\eta \\ & P_{p,t}^E / \gamma; \forall p \in \mathcal{P}, i \in \mathcal{I}^{P,+}(p), j \in \mathcal{I}^{P,-}(p), t \in \mathcal{T}; \quad (37) \end{aligned}$$

When $Q_{i,j,t,\omega}^+ > a_{p,t,\omega}^{1,-}$ and $Q_{i,j,t,\omega}^- > a_{p,t,\omega}^{2,-}$, the constraint (32) can be relaxed to the linear constraint that path the points $Q_{i,j,t,\omega}^+$ and $Q_{i,j,t,\omega}^-$ as:

$$\begin{aligned} & [P_p^Q(Q_{i,j,t,\omega}^+{}^2 + Q_{i,j,t,\omega}^+ \cdot Q_{i,j,t,\omega}^- + Q_{i,j,t,\omega}^-{}^2) + \\ & P_p^{Q,\sigma} \cdot P_{p,t,\omega}^-(Q_{i,j,t,\omega}^+ + Q_{i,j,t,\omega}^-) + P_p^\sigma \cdot P_{p,t,\omega}^-{}^2] \cdot Q_{i,j,t} \\ & - P_p^Q \cdot Q_{i,j,t,\omega}^+ \cdot Q_{i,j,t,\omega}^- (Q_{i,j,t,\omega}^+ + Q_{i,j,t,\omega}^-) \\ & - P_p^{Q,\sigma} \cdot P_{p,t,\omega}^- \cdot Q_{i,j,t,\omega}^+ \cdot Q_{i,j,t,\omega}^- \leq P_p^\eta \cdot P_{p,t}^E / \gamma; \\ & \forall p \in \mathcal{P}, i \in \mathcal{I}^{P,+}(p), j \in \mathcal{I}^{P,-}(p), t \in \mathcal{T}; \quad (38) \end{aligned}$$

B. Tightening of relaxation

The effectiveness of the proposed convex envelopes relies on the accuracy of the operational bounds $Q_{i,j,t,\omega}^+$, $Q_{i,j,t,\omega}^-$, $P_{p,t,\omega}^+$, and $P_{p,t,\omega}^-$. To improve this, we introduce a bound-tightening algorithm that iteratively adjusts these bounds for enhanced relaxation precision. Algorithm 1 provides the pseudocode for this process, which begins with broad initial bounds and refines them using a convergence tolerance ϕ and bound-tightening parameter $\epsilon_\omega \in [0, 1]$. The parameter ϵ_ω is selected to balance narrowing the bounds effectively while keeping the optimal solution within the feasible set. The algorithm solves the convexified model iteratively, adjusting bounds until the objective value converges within ϕ or the maximum iteration number is reached. This strategy refines the feasible space defined by the convex relaxation, enhancing the accuracy of the relaxation to better match the original non-convex problem, thereby improving the model's representation of true system behavior.

C. Performance Comparison of Relaxation Techniques

We compare the performance of three distinct models using a comprehensive case study. The first model, deemed the original model, adheres to constraints (1)–(24) ensures strict adherence to all constraints, offering a benchmark of solution fidelity. The second model labeled as mix-integer linear programming (MILP) introduced in [14], was used to linearize the WDN models with details provided in [9].

Algorithm 1 relaxation bound-tightening

```

1: Inputs:
    $\phi$ , sequence  $\{\epsilon_\omega\}_{\omega=1}^\Omega$ 
2: Initialize:
    $\omega \leftarrow 1$ ;  $Q_{i,j,t,\omega}^+, Q_{i,j,t,\omega}^-, P_{p,t,\omega}^+, P_{p,t,\omega}^-$ 
3: solving convexified model (1)–(5),(9)–(24),(25),(27),(29)–
   (38) to obtain  $Q_{i,j,t}, P_{p,t}, C_\omega$ ;
4:  $\omega \leftarrow 1$ 
5: while  $|C_\omega - C_{\omega-1}|/C_{\omega-1} \geq \phi$  do
6:    $Q_{i,j,t,\omega}^+ \leftarrow \epsilon_{\omega-1} \cdot Q_{i,j,t} + (1 - \epsilon_{\omega-1}) \cdot Q_{i,j,t,\omega-1}^+$ ;
7:    $Q_{i,j,t,\omega}^- \leftarrow \epsilon_{\omega-1} \cdot Q_{i,j,t} + (1 - \epsilon_{\omega-1}) \cdot Q_{i,j,t,\omega-1}^-$ ;
8:    $P_{p,t,\omega}^+ \leftarrow \epsilon_{\omega-1} \cdot P_{p,t} + (1 - \epsilon_{\omega-1}) \cdot P_{p,t,\omega-1}^+$ ;
9:    $P_{p,t,\omega}^- \leftarrow \epsilon_{\omega-1} \cdot P_{p,t} + (1 - \epsilon_{\omega-1}) \cdot P_{p,t,\omega-1}^-$ ;
10:  solving convexified model to obtain  $Q_{i,j,t}, P_{p,t}, C_\omega$ 
11:   $\omega \leftarrow \omega + 1$ ;
12: end while

```

It developed models to replace the nonlinear functions with approximation piecewise linear alternatives. The third model employs our proposed convex nonlinear programming (NLP) model in section III-A and III-B.

To assess the performance of these models, we focus on computation time and solution quality, quantified through four specific aspects. The first metric, the objective-function gap, calculates the percentage difference in the optimal objective-function value between the original model and the relaxation models (MILP and convex NLP), providing the accuracy of these relaxations in approximating true operational costs. The second metric evaluates the violation of relaxed constraints, specifically (6)–(8). We quantify the average percentage of violations across all time steps and components, defined as:

$$\begin{aligned} U^L &= 100 \sum_{l \in \mathcal{L}, i \in \mathcal{I}^{L,+}(l), j \in \mathcal{I}^{L,-}(l), t \in \mathcal{T}} [|L_l^\Delta \cdot L_l^L \cdot Q_{i,j,t}^2 - \\ & 4 \cdot \pi^2 \cdot (L_l^p)^5 \cdot \gamma \cdot H_{l,t}^L / Q_{i,j,t}^2|] / |\mathcal{T}| \cdot |\mathcal{L}|; \\ U^H &= 100 \sum_{\forall p \in \mathcal{P}, i \in \mathcal{I}^{P,+}(p), j \in \mathcal{I}^{P,-}(p), t \in \mathcal{T}} [|P_p^Q \cdot Q_{i,j,t}^2 + \\ & P_p^{Q,\sigma} \cdot Q_{i,j,t} \cdot P_{p,t} + P_p^\sigma \cdot P_{p,t}^2 - H_{p,t}^P / Q_{i,j,t}^2|] / |\mathcal{T}| \cdot |\mathcal{P}|; \\ U^E &= 100 \sum_{\forall p \in \mathcal{P}, i \in \mathcal{I}^{P,+}(p), j \in \mathcal{I}^{P,-}(p), t \in \mathcal{T}} [| \gamma \cdot H_{p,t}^P \cdot Q_{i,j,t} - \\ & P_p^\eta \cdot P_{p,t}^E / Q_{i,j,t}^3|] / |\mathcal{T}| \cdot |\mathcal{P}|; \end{aligned}$$

The third metric explores the cost implications in reality operation. By resolving the original model with these fixed pump speed values obtained from the relaxed model solutions, we gauge the feasibility and cost-effectiveness of the solutions derived from the relaxation techniques, thereby assessing their practical applicability.

IV. CASE STUDY

The proposed model is implemented on a 24-hour test 17-nodes WDN from [15] and an IEEE 24-bus PDN from [16]. The WDN consists of 17 nodes, 11 pipes, 3 VSPs, 2 tanks, and 3 junctions, along with one reservoir.

In this case, we compared the performance of three models: Original, proposed Convex NLP, and 24-segment MILP models. Both the original and proposed Convex NLP models are solved by the ANTIGONE solver and the MILP model is solved by the CPLEX solver, with default settings.

TABLE I
ESTIMATED SYSTEM-OPERATIONS COSTS (\$ MILLION), GAPS (%), AND COMPUTATION TIMES (S) IN THE CASE UNDER THREE MODELS

Model	Operation Cost	Gap	Computation Time
Original	613376.9	<i>n/a</i>	216.86
24-segment MILP	557275.9	9.15	324318.09
Convex NLP	615056.1	0.27	103.99

Table I provides a detailed comparison of the total system operations costs, the objective-function gaps between each model, and their computation times. Furthermore, Table II details the percentage of violation for each nonconvex constraint based on the solutions derived from each approximation method. In the context of a larger-scale case, our proposed convex NLP model exhibits faster computation times compared to the original model. Both the MILP model and our convex NLP model achieve low rates of constraint violations. However, the MILP model's performance in estimating system operations costs is notably less accurate and incurs considerably longer computation times. This inefficiency primarily stems from the exponential increase in the number of integer variables required as the case scale expands, complicating the computational process.

TABLE II
VIOLATIONS OF CONSTRAINTS (%) IN THE CASE UNDER TWO MODELS

Model	Constranit		
	(6)	(7)	(8)
24-segment MILP	$3e^{-3}$	$3e^{-13}$	0.42
Convex NLP	$5e^{-4}$	$7e^{-14}$	0.17

Table III reveals the practical operational costs associated with following the solutions provided by the approximation models. From the result, operating the system based on the solution from our proposed convex method incurs lower costs in practice compared to implementing the MILP solution. Those observations underscore our proposed relaxation approach's practical advantages and cost-efficiency in system operations, especially within larger-scale scenarios.

TABLE III
ACTUAL SYSTEM-OPERATIONS COSTS (\$ MILLION), AND GAPS (%) IN THE CASE UNDER TWO MODELS

Model	Operation Cost	Gap
24-segment MILP	689472.8	12.41
Convex NLP	615380.4	0.33

V. CONCLUSIONS

In conclusion, this paper addresses key challenges in WDNs by developing an advanced mathematical framework for opti-

mizing operations and reducing computation costs. Focusing on the C-OWPF, our approach leverages system interdependencies to enhance efficiency and sustainability. We tackled the nonconvex, nonlinear constraints in WDNs, particularly with VSPs, using a direct relaxation method combined with an iterative bound-tightening algorithm. This method streamlines computations and improves optimization accuracy.

Our proposed methods, validated through comparative analysis with traditional techniques, demonstrate superior computation efficiency and cost-effectiveness. They offer practical benefits, especially in larger-scale applications where traditional methods increase computational demands. This research also opens avenues for applying these optimization techniques to other interdependent infrastructure systems and more complex problem structures, inviting further innovation toward integrated operations for sustainable, efficient global infrastructure.

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